



4.5 PRACTICAL EXAMPLE

MIKROELEKTRONIKA

The most common role of a transistor in an analog circuit is as an active (amplifying) component. Diagram 4.8 shows a simple radio receiver – commonly called a “Crystal Set with amplifier.”

Variable capacitor C and coil L form a parallel oscillating circuit which is used to pick out the signal of a radio station out of many different signals of different frequencies. A diode, 100pF capacitor and a 470k resistor form a diode detector which is used to transform the low frequency voltage into information (music, speech). Information across the 470k resistor passes through a 1uF capacitor to the base of a transistor. The transistor and its associated components create a low frequency amplifier which amplifies the signal.

On figure 4.8 there are symbols for a common ground and grounding. Beginners usually assume these two are the same which is a mistake. On the circuit board the common ground is a copper track whose size is significantly wider than the other tracks. When this radio receiver is built on a circuit board, common ground is a copper strip connecting holes where the lower end of the capacitor C, coil L 100pF capacitor and 470k resistor are soldered. On the other hand, grounding is a metal rod stuck in a wet earth (connecting your circuits grounding point to the plumbing or heating system of your house is also a good way to ground your project).

Resistor R2 biases the transistor. This voltage should be around 0.7V, so that voltage on the collector is approximately equal to half the battery voltage.

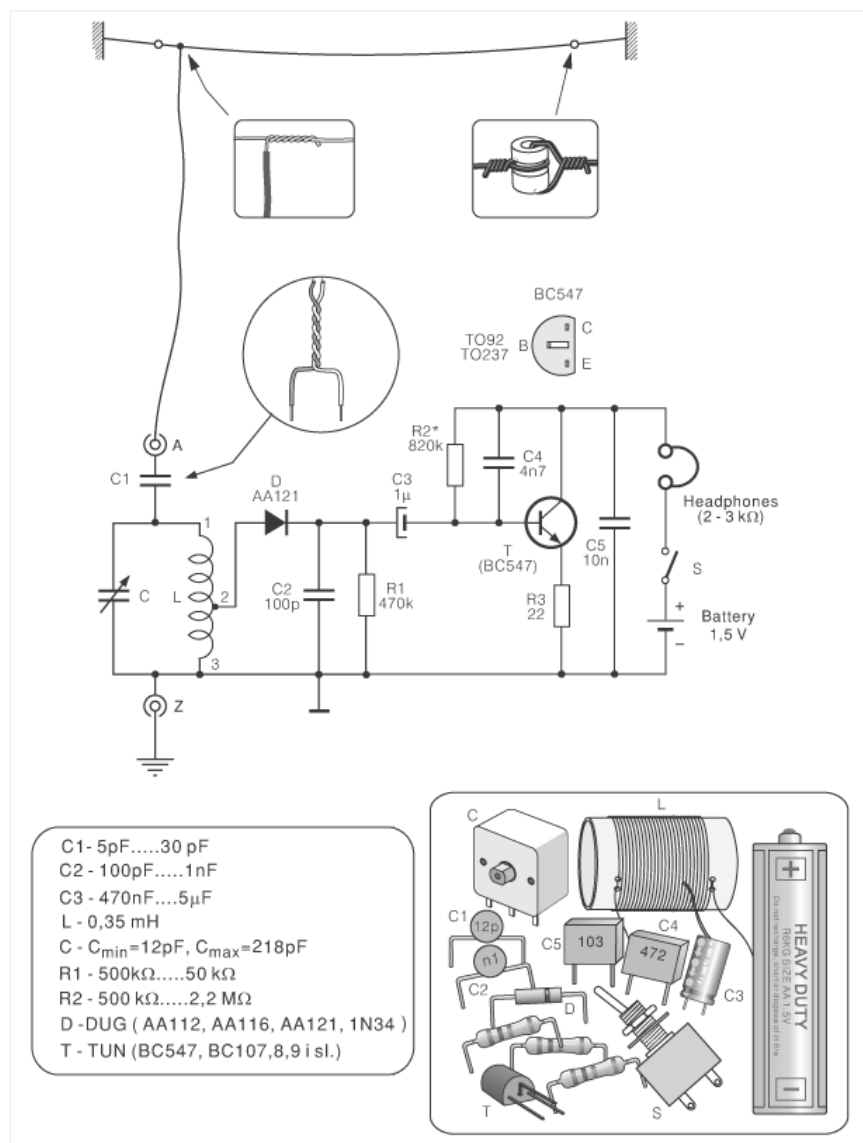


Fig. 4.8: Detector receiver with a simple amplifier



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